

Full state of the investigation

The pigeonhole principle and $AC^0[p]$ -Frege: designs, the BIKPRS route, and where it stands

Abstract

A working record. It states what is proved, what is measured, what was claimed and withdrawn, and what remains. The organising question is whether the BIKPRS reduction of $AC^0[p]$ -Frege lower bounds to Nullstellensatz degree bounds with extension polynomials can be completed. That route terminates in two lemmas about designs for the pigeonhole principle. Lemma A — a counting law for how generic linear restrictions erode a design space — is proved, uniformly in the degree, with a closed form. Lemma B — affine feasibility — is open; we record four routes to it, three of them now closed, and the single statement on which all of them converge. We also record the ancillary results: the functional pigeonhole principle has polynomial-size constant-depth refutations over every field with Boolean cofactors exhibited, with an explicit witness for why the ordinary principle does not inherit them; and a barrier analysis explaining why the measure-based approaches attempted here cannot work. A section on method errors is included because several of them were costly and structural.

Contents

1	The route	1
1.1	The chain	1
1.2	Two claims made and withdrawn	2
2	The two rings, and an error that cost a long stretch	2
3	Lemma A: the counting law	2
4	Lemma B: affine feasibility	3
4.1	Established in the correct ring	4
4.2	Counting cannot settle it	4
4.3	The four-item decomposition	4
4.4	Routes attempted and closed	4
4.5	Where the routes converge	5
5	The functional pigeonhole principle	5
6	The chessboard analysis (an account of fPHP)	5
7	Barrier analysis	6
8	Other results	6
9	What is not ours	7

1 The route

1.1 The chain

Grochow–Pitassi show constant-depth IPS over $\mathbb{F}_p$ simulates $AC^0[p]$ -Frege. BIKPRS [1] sharpen this to a degree statement: constant-depth $F(\text{MOD}_p)$ refutations of size S correspond to Nullstellensatz refutations of degree $(d_0 + \log S)(h + 1)^{O(1)}$ from the arithmetised axioms together with a leveled set of extension polynomials

$$E_{i,\bar{g}} = g_i \prod_{u \leq h} \left(1 - \sum_j r_{ju} g_j\right).$$

Their remark that “current methods apparently cannot be applied to such systems” is the obstruction attacked here.

The chain assembled:

- (1) BIKPRS: size $\rightarrow$ degree with extension polynomials. *Published.*
- (2) A design for the augmented system can be written $L(m(x)r_S) = L_x(m \cdot w_S(x))$; the extension conditions are met by $w_j = g_j^{p-2} \prod_{l < j} (1 - g_l^{p-1})$. *Verified.*
- (3) Those exact weights cost degree $(p-1)k \deg g$, k the disjunction fan-in. *Verified.*
- (4) Replacing the exact OR by the randomised $W_t = \prod_{s \leq t} (1 - (\sum_j \lambda_{sj} g_j)^{p-1})$ makes the degree $t(p-1) \deg g$, independent of the fan-in, at the cost of error p^{-t} . *Verified.*
- (5) The design must survive that error: by LP duality, PHP *augmented by t generic linear equations* has Nullstellensatz degree $> d$.

1.2 Two claims made and withdrawn

“*The chain is circular.*” False. The implications run one way and each is a theorem. What is true is weaker: both ends are degree-lower-bound statements, and degree lower bounds are dual to design construction, so the technique at the terminus resembles the technique at the start. That is a similarity of method, not an equivalence of statements.

“*Step 5 is as hard as the original problem.*” Asserted without basis, withdrawn. Toward reachability: plain Nullstellensatz degree bounds for PHP are known and design-based; the extension polynomials have $O(1)$ degree (BIKPRS Lemma 6.10); the measured thresholds grow with n ; the counting half admits closed forms. Toward difficulty: the g ’s are adversarial; counting is provably insufficient; BIKPRS reported the negative in 1997. The second list mixes properties of the object with properties of methods, and only the former is evidence. Verdict: *unknown*.

2 The two rings, and an error that cost a long stretch

Two Stanley–Reisner rings appear, and confusing them invalidated a substantial body of work.

Definition 2.1. $R = k[\Delta_{m,n}] = k[x_{ij}]/(x_{ij}x_{i'j'} : \text{exactly one of } i = i', j = j')$ — the chessboard ring, imposing **both** row- and column-exclusivity. Its monomial basis is the partial matchings. This is fPHP.

$S = k[x_{ij}]/(x_{ij}x_{i'j'} : i \neq i')$ — **column-exclusivity only**. Its monomial basis is supports with at most one cell per column, row repetition and multiplicities allowed. This is PHP.

An extended analysis was carried out in R (partial permanents, $z^{n+1} = 0$, chessboard-complex depth via BLVŽ, Schenzel's formula) and mistakenly treated as bearing on PHP. It does not. That work is recorded in §6 as a complete account of fPHP's design theory, which is what it is.

Proposition 2.2. $S = \bigotimes_{j=1}^n C$ with $C = k[y_1, \dots, y_m]/(y_a y_b : a \neq b)$; $\text{Hilb}_S(t) = (1+nt)^n/(1-t)^n$ for $m = n+1$; $\dim S = n$. Unlike R , the factorisation is complete across columns, because the hole ideal has no cross-column relations.

3 Lemma A: the counting law

Setting: $\mathcal{M}_k$ the partial matchings, $m = n+1$, $p > n$ odd, $e \perp \mu$ meaning $\text{row}(e) \notin \text{rows}(\mu)$, $\text{col}(e) \notin \text{cols}(\mu)$. For $c \in \mathbb{F}_p^{m \times n}$ and $\mu \in \mathcal{M}_{d-1}$,

$$\Phi_c(\mu) = \sum_{e \perp \mu} c_e \delta_{\mu \cup e} \in \mathbb{F}_p^{\mathcal{M}_d}, \quad H = \text{span}\{\Phi_{R_i}(\mu)\},$$

R_i the row- i indicator, and r_s the rank of $\{\Phi_{c^s}(\mu)\}_\mu$ modulo $H + \sum_{s' < s} \text{span}\{\Phi_{c^{s'}}(\mu)\}$.

Theorem 3.1. For n large relative to d and $p > n$ odd,

$$r_s = |\mathcal{M}_{d-1}| - \left(\text{def}(d) + (s-1)\text{cost}(d) - |\mathcal{M}_{d-3}| \binom{s-1}{2} \right),$$

with $P(n, r) = n(n-1) \cdots (n-r+1)$ and

$$\text{def}(d) = \binom{m}{d-1} \sum_{k=1}^{d-1} (-1)^{k-1} \binom{d-1}{k} P(n, d-1-k), \quad \text{def}(1) = 0,$$

$$\text{cost}(d) = |\mathcal{M}_{d-2}| - \text{def}(d-1), \quad R(d) = |\mathcal{M}_{d-3}|.$$

Fibrewise antisymmetry. For $\mu' \in \mathcal{M}_{d-2}$ and any c, c' , $\sum_{e \perp \mu'} c_e \Phi_{c'}(\mu' \cup e) = \sum_{e \perp \mu'} c'_e \Phi_c(\mu' \cup e)$: the coefficient of δ_ν , $\nu = \mu' \cup \{e_1, e_2\}$, is $c_{e_1} c'_{e_2} + c_{e_2} c'_{e_1}$ on both sides. Equivalently $L_c L_{c'} = L_{c'} L_c$, so this holds in *any* commutative ring and in particular transfers to S .

Row identity. $\Phi_{R_i}(\mu) = H_i(\mu)$. Substituting $c = R_i$ gives $D_{i, \mu'} : \sum_{e \in \text{row } i, e \perp \mu'} \Phi_c(\mu' \cup e) \in H$, with $\beta(k, \mu') = [k=i] \sum_{f \in \mu'} c_f$. The *sum* weight works at all d ; the *product* weight $\prod_{f \in \mu'} c_f$ works only at $d=3$. Verified 80/80, 150/150, 252/252.

Full rank of the λ family. If $\sum_{\mu'} \alpha_{\mu'} \lambda^{\mu'} = 0$ then $\sum_{e \in \mu} \alpha_{\mu \setminus e} c_e = 0$ for all μ . Order cells lexicographically, pick for each μ' a compatible $e(\mu')$ exceeding all its cells, take μ' maximal with $\alpha_{\mu'} \neq 0$ and evaluate at $\mu' \cup e(\mu')$: all other terms have larger index, hence vanish, so $\alpha_{\mu'} c_{e(\mu')} = 0$. No base case needed; transfers to S .

$\text{def}(d)$. The vectors $\lambda^{(i, \mu')}$ form the incidence matrix of a $(d-1)$ -uniform hypergraph: each $\mu \in \mathcal{M}_{d-1}$ lies in exactly $d-1$ of them. The pigeon set $\text{rows}(\mu') \cup \{i\}$ is invariant along edges, so components are indexed by $(d-1)$ -subsets of $[m]$. Within a component, edges are injective maps $S \rightarrow [n]$ and vertices their restrictions omitting one point; the row space is the span of functions ignoring one coordinate, of dimension $\sum_k (-1)^{k-1} \binom{d-1}{k} P(n, d-1-k)$ by inclusion-exclusion. Exact at $(6, 3)$ and $(7, 3)$: 231 and 364.

cost(d): the level shift. Written out,

$$\text{def}(d) = \dim\{\lambda : \exists \gamma, \sum_{e \in \nu} \lambda_{\nu \setminus e} c_e = \sum_{e \in \nu} \gamma(i_e, \nu \setminus e) \forall \nu \in \mathcal{M}_d\},$$

while admissibility of α at level d is the same system with $d \rightarrow d - 1$. Hence $\dim\{\text{admissible}\} = \text{def}(d - 1)$ and $\text{cost}(d) = |\mathcal{M}_{d-2}| - \text{def}(d - 1)$, with no substitution.

$R(d) = |\mathcal{M}_{d-3}|$. *Lower:* for $\mu'' \in \mathcal{M}_{d-3}$ put $\alpha_{\mu'}^{\mu''} = [\mu'' \subset \mu'] c'_{\mu' \setminus \mu''}$, $\beta^{\mu''}$ likewise with c ; with $\mu \setminus \mu'' = \{g, h\}$ both give $c'_h c_g + c'_g c_h$, symmetric. Independence from injectivity of T_c plus full rank one level down. *Upper:* T_c is multiplication by L_c on $k[\Delta_{m,n}]$ and $T_c T_{c'} = T_{c'} T_c$; the bound says $L_c, L_{c'}$ is a regular sequence, which follows from $\text{depth } k[\Delta_{m,n}] \geq \nu_{m,n}$ [2, Cor. 1.4] with $\nu_{m,n} = \min\{m - 1, \lceil \frac{m+n-4}{3} \rceil\}$ [3], giving $\approx 2n/3$ at $m = n + 1$ — far more than the two forms needed.

Confirmation. 16/16 values of r_s ($d = 2, 3; n = 5, 6, 7$); 9/9 values of $R(d)$ ($d = 2, 3, 4; n = 5, 6, 7$, including $n = 7, d = 4$ giving $56 = |\mathcal{M}_1|$); def exact where $d \leq n/2$ and deviating exactly where that fails.

Remark 3.2 (saturation). Designs for PHP_n^{n+1} exist precisely for $d \lesssim n/2$ (Razborov, via duality). This explains every saturation observed and removes what had been flagged as the main uniformity risk: the chain needs $d = \Theta(\log S) = \text{poly } \log n$ against a ceiling of $n/2$.

4 Lemma B: affine feasibility

Problem 4.1 (Lemma B). With $\bar{B} = S/(\rho_i - \rho_1)$, $\rho_i = \sum_j x_{ij}$, $z = \rho_1$, determine

$$t_{\max}(d) = \max\{t : z^d \notin (L_{c^1}, \dots, L_{c^t})\bar{B}\}$$

for c^s arising as BIKPRS extension data, in the regime $d = \Theta(\log S)$, $n \rightarrow \infty$; specifically show $t_{\max}(d) = \Omega(n)$.

4.1 Established in the correct ring

Proposition 4.2. $z^k = \rho_1 \cdots \rho_k = P_{[k]}^{[n]} = \sum_{f: [k] \hookrightarrow [n]} \prod_{i \leq k} x_{i, f(i)}$. Each factor contributes a distinct row, so no same-row product arises; column collisions vanish. Hence $z^{n+1} = 0$ and $z^n \neq 0$, and $\max\{k : z^k \neq 0\} = n$, matching $\deg_{NS}(\text{PHP}_n^{n+1}) = n + 1$ for $p > n$ through duality.

Proposition 4.3. $\dim \bar{B} = n - 1$; the $\rho_i - \rho_1$ are not a system of parameters. On the component where all columns use one row, the relations degenerate.

Proposition 4.4 (injectivity in S). For $j < n$ and any row i , $\times \rho_i : S_j \rightarrow S_{j+1}$ is injective. For $\mu \in S_j$ let $F_i(\mu)$ be the columns free or used by μ in row i ; $|\text{cols}(\mu)| \leq j < n$ so $F_i(\mu) \neq \emptyset$. Then $\rho_i \mu = \sum_{l \in F_i(\mu)} x_{il} \mu$, and with $l^* = \max F_i(\mu)$ one has $F_i(x_{il^*} \mu) = F_i(\mu)$, so $\mu \mapsto x_{il^*} \mu$ is injective on monomials and supplies distinct leading terms under a suitable graded lex order.

4.2 Counting cannot settle it

At $(n, d) = (6, 3)$ the design system becomes infeasible at $t = 3$ while 966 dimensions of homogeneous freedom remain. Hence no rank or dimension argument — including Theorem 3.1, however sharp — decides Lemma B.

4.3 The four-item decomposition

statement	status
(a) $\dim \bar{B}_d = \binom{n}{d} n^d$ for $d < n$	equivalent to $H_i(\rho; S)_j = 0$ for $i \geq 1, j < n$; measured 5/5; unproved
(b) z^d dies exactly at exhaustion of $\bar{B}_d$	measured 2/2; unproved; carries the content
(c) the erosion law	relation family and full rank proved in S ; counting formula measured
(d) uniformity over adversarial c^s	untouched

With (a)–(c) one gets $t_{\max}(d) = \Theta(n^2/d)$ for *generic* forms, which at $d = \text{poly log } n$ is $\tilde{\Theta}(n^2)$ — far above the $\Omega(n)$ required. (d) is independent: the BIKPRS c^s are coefficients of $\sum_j r_{ju} g_j$, not generic, and no generic statement implies an adversarial one.

4.4 Routes attempted and closed

- (i) *Weak Lefschetz*. WLP for $\bar{B}$ would give (a), (b), (c) at once. But S is a tensor product of $n \geq 3$ quadratic factors, and tensor products with more than two such factors are known to fail WLP, with tautological kernel elements [4]. Not available.
- (ii) *The coefficient-sum functional* $\Lambda_d(\nu) = [\nu \in \text{supp}(z^d)]$. Then $\Lambda_d(L_c \mu) = \sum_{e \perp \mu, \text{row-new}} c_e$, and requiring this to vanish for all $\mu \in \mathcal{M}_{d-1}$ forces $c = 0$ by Möbius inversion. So Λ_d annihilates no nonzero restriction, structured or generic.
- (iii) *Pure powers*. For $\lambda = \ell^d$ with ℓ of column-injective support, some row is untouched, so its row sum vanishes and $\lambda(z^d) = d! \prod_{i \leq d} (\sum_j \ell_{ij}) = 0$. Every pure power pairs to zero with z^d ; a witness must be a cancelling sum.
- (iv) *Symmetric functionals*. No $S_m \times S_n$ -symmetric λ annihilates a generic L_c : the condition collapses to $\sum_{i \notin \text{rows } \mu} \sum_{j \notin \text{cols } \mu} c_{ij} = 0$ for all μ , false for generic c . Five explicit families (symmetric, product, determinantal, LGV path-matrix, linear-in- μ) were refuted against Razborov’s design as a control, which passed 0/445 while four of the five gave 433/445 violations.

4.5 Where the routes converge

At $d = 2$, a functional is a symmetric bilinear form B on $\bar{B}_1$ with $B(x_{ij}, x_{i'j}) = 0$ (same column), $L_{c^s} \in \text{rad } B$, and $B(z, z) \neq 0$. Now $B(z, z) = \sum_{j,j'} B(x_{1j}, x_{1j'})$ involves only **same-row** pairs, while the constraints involve only **same-column** pairs — and in S , unlike R , same-row pairs carry no relation. So $B(z, z) = 0$ is not directly imposed; the question is whether the t restriction-identifications *propagate* to kill it. That single propagation question is (b) at $d = 2$, and the general- d case has the same shape: the constraints are column-only, the target row-only.

Two independent counts agree on the threshold: exhaustion of Sym^2 against the column constraints gives $t \lesssim n^2 - n^{3/2}$, and the erosion law gives the same, matching the measured $t_{\max}(2) = 3$ at $n = 3$.

5 The functional pigeonhole principle

Theorem 5.1. *Over any field, fPHP $_n^m$ with $m > n$ has a refutation of size $O(m^2 n^2)$ and constant depth, with Boolean cofactors exhibited.*

Proof. Let $\Phi(t) = \prod_{i,j}(1+tx_{ij})$. Row-wise $\prod_j(1+tx_{ij}) = 1+t\rho_i + \sum_{k \geq 2} t^k e_k(\text{row } i)$, each e_k with $k \geq 2$ a sum of row-exclusivity axioms; using $\rho_i = 1$, each row factor is $\equiv 1+t$, so $\Phi \equiv (1+t)^m$. Column-wise each $e_k(\text{col } j)$, $k \geq 2$, is a sum of hole axioms, so $\Phi \equiv \prod_{j \leq n}(1+t\sigma_j)$, of degree $\leq n$ in t . Comparing t^m coefficients gives $1=0$. Cofactors are products of the remaining factors; extracting the t^m coefficient is interpolation at $m+1$ points. $\square$

The generating function is Ben-Or's construction [5, Thm. 1.15]; what is specific is the two-sided reading and the resulting cofactors.

Proposition 5.2 (why PHP does not inherit it). $\prod_j(1+tx_{ij}) - (1+t\rho_i) = \sum_{k \geq 2} t^k e_k(x_{i1}, \dots, x_{in})$, so the proof transfers exactly when $e_k(\text{row } i) = 0$ for $k \geq 2$. Over $\mathbb{F}_p$ these fail: a row with $p+1$ ones has $\rho_i \equiv 1$ but $e_p \equiv 1 \neq 0$. Over $\mathbb{F}_3$ a row of four ones gives $e_1 \equiv 1$, $e_2 \equiv 0$, $e_3 \equiv 1$.

Every easiness construction attempted for PHP — the Fermat product, the counting certificate $\prod_{k \leq n}(S-k)$, inheritance from fPHP, the generating function without row-exclusivity, and $y_{ij} = x_{ij} \prod_{j' < j}(1-x_{ij'})$ — fails at exactly this point: $\rho_i = 1$ over $\mathbb{F}_p$ does not say “exactly one”.

6 The chessboard analysis (an account of fPHP)

Recorded as what it is. $R = k[\Delta_{m,n}]$, $\bar{A} = R[z]/(\sigma_i - z) \cong R/(\sigma_i - \sigma_j)$ with $z \mapsto \sigma_1$.

- $z^k = P_{[k]}^{[n]}$; $z^n = \text{perm}$ ($n!$ terms); $z^{n+1} = 0$, which is the pigeonhole principle as a monomial identity.
- $\bar{A}$ is Artinian: the $\sigma_i - z$ are $m = \dim R[z]$ forms with $V = \{0\}$, the rows properly colouring Δ .
- $\text{Ann}_R(z^d)$ is the monomial ideal $\{x^a : |\text{cols}(a)| - |\text{rows}(a) \cap [d]| > n-d\}$ — itself a pigeonhole condition on the residual board.
- $\Delta_{n+1,n}$ is a closed orientable $(n-1)$ -manifold (induct on links, $\text{lk}(v) = \Delta_{n,n-1}$, base $\Delta_{2,1} = S^0$), hence R is Buchsbaum; Schenzel's formula gives $\dim \bar{A}_j$ from the h -vector and the Betti numbers, and $\dim \bar{A}_j = h_j$ for $j \leq \nu$.
- Hilbert function of $R/(\text{row sums})$: $[1, 8, 6]$, $[1, 15, 50, 10]$, $[1, 24, 165, 280, 15]$, $[1, 35, 399, 1645, 1470, 21]$ for $n = 3, \dots, 6$; proved $H(1) = n^2 - 1$, $H(j) = 0$ for $j \geq n$; observed $H(n-1) = \binom{n+1}{2}$.

Remark 6.1. Ann was described as the object Lemma B needs. That was wrong: B is an *ideal-membership* question ($z^d \in (L_{cs})$), not an *annihilation* question. The two are distinct and the annihilator does not settle membership.

7 Barrier analysis

Proposition 7.1 (few-valued axioms). *The functional half of a refutation is free if either the field is $\mathbb{F}_p$ and the f_j are encodable — $[f=0] = 1 - f^{p-1}$, and $G_j = f_j^{p-2} \prod_{i < j}(1 - f_i^{p-1})$ telescopes to 1 — or each f_j takes $O(1)$ values, where $G_j = \prod_{i < j}(1 - f_i)$ works over every field with integer coefficients.*

Known [6]; derived independently here. The certificate uses no Frobenius and is integral, ruling out p -adic, Witt-vector and prismatic obstructions.

Theorem 7.2 (closure). *A certificate class closed under products contains the free certificate, of size $O(M^2)$ and constant depth. Hence any measure with a class bound has $\min_{\text{coset}} \mu \leq \Phi(\text{poly})$, and no superpolynomial IPS'_{LIN} bound follows.*

Calibration: for PHP over $\mathbb{F}_3$ the free certificate has constant depth but *full* multilinear degree (6/6 at $n = 2$, 12/12 at $n = 3$), so it is excluded exactly by the multilinear and bounded-individual-degree restrictions — the classes where bounds are known and which the CNF translation cannot afford.

Proposition 7.3. *Every CNF instance has $\text{ind} = 0$: a clause arithmetises to a fixed product with coefficients in $\{-1, 0, 1\}$.*

Every published constant-depth IPS bound is proved on an instance with free coefficients; the translation requires CNFs; these are incompatible, and gadget variables added to the encoding become variables of the CNF, changing the tautology.

The coll/ind framework. Easiness needs coll small, hardness needs ind large. Correct on all four instances with settled status (subset-sum, symmetric multilinear: easy; lifted subset-sum: hard; PHP: open). PHP has coll large by three independent obstructions and $\text{ind} = 0$, and the two are antagonistic: weighting the pigeon axioms raises ind but drops the Nullstellensatz degree to 1.

Featurelessness. PHP’s matroid is uniform $U_{n,m}$, representable over every field; its contextuality is maximal with threshold n ; under single-coefficient perturbation it is a local maximum of coset dimension only because perturbations move it to a lower degree stratum, and random systems at matched axiom count reach strictly more. Every structural invariant computed is a function of (m, n) .

8 Other results

Proposition 8.1. *For $p \mid (n+1)!n!$ and σ a p -cycle on pigeons, PHP_n^{n+1} over $\mathbb{F}_p$ admits certificates but no σ -invariant certificate. Verified $n = 2, 3$; survives restriction to a Sylow p -subgroup and to $\langle \sigma \rangle$; vanishes on p' -subgroups. The obstruction is thin: the syzygy module is 95.7% free at $n = 2$, 99.1% at $n = 3$.*

Proposition 8.2. $\deg_{NS}(\text{PHP}_n^{n+1}) \leq n + 1$ over any field, with equality over $\mathbb{F}_p$ for $p > n$. Over $\mathbb{F}_3$: 3, 3, 4 for $n = 3, 4, 5$. Over the algebraic closure: 3, 4, ≥ 5 for $n = 2, 3, 4$.

Proposition 8.3. $\det(XX^T) = 0$ identically and $XX^T \equiv \text{diag}(p_i)$ modulo the hole ideal, giving $\prod_i p_i \in I$ with signed-minor cofactors. Neither transfer works: the Gram variety has $\dim = mn - \binom{n}{2}$ against $(m - 1)^2$ needed, failing for $n \geq 4$; and IMM is excluded under every partition, since 63 of 84 cofactor monomials carry a repeated variable. Squaring X supplies the rank deficiency and destroys set-multilinearity simultaneously.

9 What is not ours

Independently derived here but published: the free certificate and few-valuedness [6, “crucial observation”, Lemma 3.1]; the localisation of difficulty in the Boolean cofactors [6, Remark 1.8], with the same $xy = 2$ example; Lucas digits, Ben-Or’s e_k construction, constant-depth multilinearisation, and easiness of symmetric multilinear systems [5]. Our $\Phi(t)$ is Ben-Or’s construction.

Corrections to our own claims: LST bound IMM, not the determinant, and need characteristic 0 or large — but Forbes (CCC’24) extends to any field, so the characteristic objection was itself too strong. Lu–Santhanam–Tzameret [7] obtain unconditional $AC^0[p]$ -Frege bounds by diagonalisation

on self-referential families, so the blanket dismissal of diagonalisation was wrong; what remains true is that it does not reach natural principles, the family having to be constructed against the proof enumeration.

10 Open problems

- B1.** *The propagation question.* At $d = 2$: for generic c^s below exhaustion, do the restriction-identifications force $B(z, z) = 0$? Constraints are same-column, the target same-row, and in S these are disjoint. This is (b), and all routes converge on it.
- B2.** *Koszul vanishing.* $H_i(\rho_1, \dots, \rho_m; S)_j = 0$ for $i \geq 1, j < n$. Equivalent to (a) and to injectivity on $\bar{B}$; the single-form case is proved above. The first obstruction must sit at degree $n + 1$, where $\rho_1 \cdots \rho_m = 0$.
- B3.** *Adversarial uniformity.* (d). Untouched, and not implied by any generic statement.
- B4.** *Verification at $d = 4$* of Theorem 3.1’s full r_s sequence; the prediction at $n = 6$ is second difference $R(4) = |\mathcal{M}_1| = 42$.

11 Method errors

Recorded because they were structural and costly.

- *The ring confusion.* An extended analysis was carried out in the chessboard ring R and treated as bearing on PHP. R imposes row-exclusivity; PHP does not. The tell — BLVŽ’s Stanley–Reisner presentation, which lists same-row products — was quoted approvingly while checking a citation and read as confirmation. The same error then recurred inside a purported proof of an upper bound on $t_{\max}$, whose three separate flaws were one flaw: the argument was valid in R and written in S .
- *A false measurement.* A run reporting $t_{\max} = 0$ was produced by `{(i,j): Random(seed*1000+s).randrange for ...}`, which reseeds inside the comprehension and assigns every cell the same value; the “generic” forms were all multiples of z . Three controls passed because none exercised the form construction. It was caught by proving that the reported behaviour (rank saturation) would force $\dim \bar{B}_d \leq \dim \bar{B}_{d-1}$, which is false.
- *Annihilator versus membership.* B was described as reducing to $\text{Ann}(z^d)$. It does not: B is ideal membership.
- *Overstated framing.* “Reduces to”, “last step”, “circular” were each used for what were equivalences or one-directional reductions of unknown relative difficulty.
- *Controls.* Several retractions followed from controls that compared an object to itself, compared ranks across separate pivot assignments, requested designs at or above the Nullstellensatz degree, or used a degree budget so small that all constraints were silently discarded.
- *Vectorisation.* A first “vectorised” eliminator using `np.unique` and `np.add.at` in the inner loop was slower than dictionaries. The working core switches adaptively between sparse rows with `searchsorted` merges and dense rank-one updates, and is validated against a dense reference.

References

- [1] S. Buss, R. Impagliazzo, J. Krajíček, P. Pudlák, A. Razborov, J. Sgall, *Proof complexity in algebraic systems and bounded depth Frege systems with modular counting*, Comput. Complexity 6 (1996/97).

- [2] A. Björner, L. Lovász, S. T. Vrećica, R. T. Živaljević, *Chessboard complexes and matching complexes*, J. London Math. Soc. (2) 49 (1994), 25–39.
- [3] J. Jonsson, *On the 3-torsion part of the homology of the chessboard complex*, arXiv:1203.5644.
- [4] M. Michałek, R. M. Miró-Roig, *Smooth monomial Togliatti systems of cubics*; and T. Q. Hoa, N. D. Phuoc, T. N. T. Son, *The weak Lefschetz property for tensor products of Artinian monomial algebras*, arXiv:2510.21569.
- [5] A. Behera, N. Limaye, V. Ramanathan, S. Srinivasan, *New bounds for the Ideal Proof System in positive characteristic*, ICALP 2025; arXiv:2506.16397.
- [6] A. Behera, K. A. Hansen, N. Limaye, S. Srinivasan, *Separation results for constant-depth and multilinear ideal proof systems*, arXiv:2601.06299.
- [7] Z. Lu, R. Santhanam, I. Tzameret, *$AC^0[p]$ -Frege cannot efficiently prove that constant-depth algebraic circuit lower bounds are hard*, ECCC TR25-134.